



Princeton Computer Science Contest – Fall 2025

Problem 2: Cell Capture (10 points) [Codeforces]

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It is important to realize that in physics today, we have no knowledge of what energy is.

– Richard P. Feynman

Deep in Jadwin Hall, a team of researchers has been experimenting with a new type of particle that appears to gravitate towards certain energy cells on a metal plate. The plate consists of m rows and n columns of cells, each having a certain energy level.

In an experiment, a particle can start from any location. Once it occupies a cell, it may attempt to move to a neighboring cell to capture it. However, its movement is governed by the laws of *energy amplification*. Specifically, the particle can move from its current cell into an adjacent one only if that cell's energy level is **strictly higher** than the one it currently occupies (since the particle only seeks to increase its energy state).

Motion is limited to directly adjacent cells (north, south, east, or west). The particle cannot move diagonally and does not leave the plate since that would take it to a lower energy state.

However, the researchers are worried that the particle may capture too many cells, causing the plate to become unstable. Help them determine the maximum number of cells that the particle can capture during a single experiment, assuming it always follows the rule of moving to higher energy.

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Input

The first line contains two integers m and n — the number of rows and columns of the metal plate. The following m lines each contain n integers, representing the energy levels of the cells.

- $1 \leq m, n \leq 234$
- Each energy level is a non-negative integer not exceeding the maximum value of a 32-bit signed integer.

Output

Print a single integer corresponding to the maximum number of cells the particle can capture during a single experiment.

Example

Input:

```
3 3
10 10 5
7 7 9
3 2 2
```

Output:

```
4
```

Explanation: In the sample case above, the longest sequence of valid captures has length 4, such as the path $[2, 3, 7, 10]$.

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